

K'Nex't Level Engineering

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Introduction.....	4
Literature Review	4
1. Relevant Theory.....	4
1.1 Roller Coaster Terminology	4
1.2 K’Nex Terminology.....	4
1.3 Relevant Newtonian Mechanics Concepts	5
2. Scaled Roller Coaster Information	6
2.1 Jared Pincus Research Paper	6
2.2 Jon Mendenhall Model	6
3. K’Nex Modifications Community	7
3.1 The K’Nex Makers Discord	7
3.2 The K’Nex Enthusiast	7
3.3 Clover Coasters	7
4. Education Techniques	7
4.1 Computer-Based Labs	7
4.2 K’Nex in Education.....	8
5. Filament Extruder Assembly Components	8
Initial Designs	8
1. Driving Mechanisms.....	8
1.1 Rack and Pinion.....	8
1.2 Friction Wheel	9
1.3 Filament Extruder Mechanism	9
2. Filament Extruder Wheel Assembly	9
2.1 Smaller Tooth Gear and U-Shaped Idler (Stepper Motor Test).....	10
2.2 Smooth Drive Wheel and Idler	10
2.3 Toothed Drive Wheel and U-Shaped Idler Wheel Assembly	11
2.4 Front Cluster Side-Guide Wheel Addition (With Spring)	11
2.5 Centered Side-Guide Wheel	12
2.6. Crown-Structured Wheel Cluster	12
2.7. Swivel Cart	13
3. Top Shell Design.....	14
3.1 Model Seats	14
3.2 Battery Holster.....	15
4. Modelling.....	15
4.1 MATLAB Code	15
4.2 MATLAB Friction	17
4.3 Scale Factor	17
5. Control Mechanisms	18
5.1 Open Loop System	18
5.2 Closed Loop System.....	18
5.3 Bridged System	18

5.4 Control Mechanism Implementation	19
Decision Criteria	20
1. Driving Mechanisms.....	20
1.1 Filament Extruder Wheel Assembly.....	20
1.2 Top Shell Design.....	21
2. Control Mechanisms	21
3. Tracker Software Analysis	21
Final Design.....	22
1. Cart Design.....	22
2. Controls Design.....	23
Analysis & Testing.....	24
1. Track Construction.....	24
1.1 Prototype Track	24
1.2 Final Track	25
2. Tracker Analysis	27
2.1 Tracker Familiarization and Initial Software Testing	27
2.2 Experimental Procedure for Baseline Cart Testing	28
2.3 Initial Test Results and Data Analysis	29
2.4 Final Analysis of unmodified and motor-controlled cart on 2D track.....	31
3. MATLAB Integration	33
3.1 Calibration	33
3.2 Control System	33
4. Motor Testing (Centered Side Guide Wheel)	34
4.1 Test Overview	34
4.2 Test Results.....	34
Summary and Next Steps.....	35
References	37
Appendix A	39
Final Cart Design Images	39

Introduction

Many middle and high school science classes use K’Nex models to teach motion and energy concepts. The goal of this project is to rectify the misrepresentation of rollercoaster train motion in the K’Nex STEM Explorations: Roller Coaster Building Set [1]. The misrepresentation is due to the uneven scaling of kinematics on dimensionally scaled gravity driven rollercoaster models. K’Nex Education Sets are advertised to “allow students to build replicas of real-world machines and other models [2].” Dimensional analysis (see Relevant Theory section) shows that the model does not scale with gravity evenly, producing the visual discrepancy. By improving the train’s motion to better represent its full-size counterpart, this project will increase the educational value of this set.

Literature Review

1. Relevant Theory

1.1 Roller Coaster Terminology

Roller coaster wheel assemblies were reviewed to establish consistent terminology for use throughout this project. Based on this review, standardized terminology was adopted, as shown in Figure 1 [3]. The complete wheel assembly of a roller coaster is referred to as a wheel cluster. Individually, the top wheels are referred to as road wheels, the side wheels as guide wheels, and the bottom wheels as up-stop wheels. The road wheels tend to be the largest and are replaced most frequently as they are under the highest load from constant contact with the track.



Figure 1: Defined wheel cluster components [3]

A roller coaster train consists of multiple carts, while the track consists of rails supported by structural framing. The rails interact with the wheel clusters of the carts.

1.2 K’Nex Terminology

To further aid communication of designs, K’Nex pieces were researched for proper naming. The first useful definition were the types of K’Nex sets. Classic K’Nex and Micro K’Nex are the two sets of interest [4]. While similar in shapes, the size of classic and micro K’Nex are different.

Within each set, there are two main types of pieces: Rods and Connectors. Rods describe the long bars and connectors describe the joining pieces between them.

Rods come in varying lengths; the classic set has 6 standard lengths and the micro set has 7 standard lengths [5]. Table 1 displays the lengths of the rods as organized by each set type below.

Table 1: K'Nex Rod Lengths

Classic	Micro
-	14mm
16mm	25mm
32mm	40mm
54mm	63mm
86mm	94mm
120mm	138mm
190mm	200mm

Connectors vary based on the number of connection points, ranging from 2 to 8 connections points. With a maximum connector having 8 connection points, the points are spaced in 45° increments. One outlier of this spacing standard is the 180° 2-way connector that connects rods while keeping them axially aligned. To allow for 3-Dimensional (3D) assemblies, 4-way and 7-way connectors come in variations with a slot to connect two connectors perpendicularly [6]. When assembled in this manner, these connectors form a 3D connector.



Figure 2: Summary of K'Nex Component Types [5]

1.3 Relevant Newtonian Mechanics Concepts

Roller Coaster train motion involves a range of physics concepts, but the most direct categories include energy conservation and kinematic equations. Energy conservation equations are used to describe the exchange between potential and kinetic energy within a system. This transfer of energy results in changes in height and speed of the train. The unmodified K'Nex train and standard full-scale trains are gravity driven; without the addition of energy in these cases, the mathematical representation simplifies down to potential, kinetic, and dissipative energy loss (friction). Friction loss is inconsistent between the K'Nex model and full-scale trains which adds to the uneven scaling of gravity-driven coasters. Introducing a motor (a driving energy source) allows us to adjust the K'Nex train's motion to account for the uneven kinematic scaling as seen through potential, kinetic, and dissipative energy. It should be noted that for controlling methods the ideal system (no frictional losses) is sufficient for scaling. As the changes in kinematics are aimed at visual consistency not perfect modeling, small discrepancies from neglecting friction still meet the visual goal.

A focus on energy transfer is beneficial for conceptual understanding and education; a focus on kinematics adds detail and is the better theory to control the model train's motion. This focus involves measuring the position, velocity, and acceleration of the train along the track. Kinematics introduce vectors where each value has a magnitude and a direction while energy only describes a magnitude. Although gravitational forces are vector forces, the velocity is

proportionally related to kinetic energy which is a scalar value. The presence of this scalar value (Kinetic Energy) simplifies modeling calculations as the direction of travel can be neglected in related scaling calculations.

2. Scaled Roller Coaster Information

2.1 Jared Pincus Research Paper

In a Bridges 2024 Conference Paper, Jared Pincus, a Boston PhD student, discussed the issues and solutions involving the scaling of model roller coasters [7]. Focused on energy transfer as summarized above, Jared derives the ideal relationship – meaning this derivation ignored friction and other dissipative forces – between velocities of model roller coaster carts and their real-world counterparts in this paper; shown below in Equation 1:

$$\frac{v_1}{v_2} = \sqrt{\sigma} \quad (\text{eq. 1})$$

where V_1 is the model velocity at the bottom of a hill, V_2 is the real-world velocity at the bottom of a hill, and σ is the ratio of the model hill height to the real-world hill height. σ can also be referred to as the size scale factor. Jared acknowledged that this relationship is incomplete and is dependent on significant assumptions but still describes the fundamental flaw with gravity-driven model roller coasters. Importantly, this incomplete model is still sufficient for creating visual accuracy and can be used for controlling the altered cart. While model coasters change in size by a factor σ , model coaster velocities only change by a factor $\sqrt{\sigma}$; this results in uneven scaling of variables.

2.2 Jon Mendenhall Model

To address the limitations of gravity-driven model coasters, Jon Mendenhall designed a visually accurate model roller coaster shown in Figure 3 below [8].



Figure 3: Jon Mendenhall's Accurately Scaled Model Roller Coaster [8]

This roller coaster used a rack and pinion mechanism shaped to the layout of the track. This model was custom built and 3D printed; the cart's onboard motor and driving gear are controlled using code generated by Mendenhall. Although this design resolves the educational inaccuracies of the K'Nex STEM Explorations set, its highly customized fabrication and programming requirements limit accessibility for broader educational use. Students learning foundational

physics concepts cannot reasonably be expected to replicate such a complex system, reinforcing the need for a more accessible and reproducible solution, motivating our project.

3. K’Nex Modifications Community

3.1 The K’Nex Makers Discord

The K’Nex Makers is a YouTube channel outlining the creation of custom 3D-printed trains for Classic K’Nex component coasters [9]. The STEM Explorations Set uses Micro K’Nex which presents similar but different issues. Since the two sets share similar geometries, building strategies and other lessons can still be taken from this channel. The integration of 3D Printed components at this scale is useful as the channel outlines tips for train and wheel assembly design. Additionally, many track designs presented by the channel provide insight into effective use of K’Nex as a structural medium, including support structure strategies.

Additionally, the creators of the channel host a community discord [10]. This discord channel supports communication between K’Nex coaster hobbyists for a range of topics. Specific discussions have not yet been used by our team, but the research on this channel provides a location for K’Nex technical experts in future consulting.

3.2 The K’Nex Enthusiast

The K’Nex Enthusiast YouTube channel is more directly applicable as many projects by this creator include Micro K’Nex [11]. Beyond personal projects of recreating model versions of real-world coasters, this channel contains tutorials for building track spines and support structures. With detailed recommendations regarding building methods, our team can more easily use Micro K’Nex as an effective medium for designing testing structures (See Testing and Analysis for more). It should be noted that this user is much smaller in views, so a community is not as readily available, however, the video content contains significant educational merit.

3.3 Clover Coasters

Clover Coasters is an active channel that recently released a 3D printed train that integrates with Micro K’Nex [12]; this design is now for sale [13]. The use of many small screws and effective placement of bearings to act as wheels highlights the importance of a detailed wheel assembly design. The many small components allow for the wheel assembly to move effectively with the track to reduce misalignment and consequential energy loss. To further reduce frictional energy loss, all wheels in this design are ball bearings. This design is not fit for a driving mechanism but is a highly efficient gravity-driven train. Due to the high craftsmanship and videos detailing the design process and final product, this channel is a great resource for future wheel assembly design. Like the K’Nex Enthusiast, this creator is relatively new and not well known.

4. Education Techniques

This project aims to increase the educational value of the K’Nex STEM Explorations Set [2]; the following research depicts current educational techniques related to amusement parks and physics concepts.

4.1 Computer-Based Labs

Digital interactive lessons have been developed to avoid the arduous task of adjusting model roller coasters to demonstrate variables related to the train’s motion (Track Layout, Frictional Factors, and Train Construction) [14]. While these tools are stronger in their ease of educational

display, they lack the hands-on feature that model coasters add. Hands-on education has been shown to have a significant benefit in engagement and learning benefits [15].

4.2 K'Nex in Education

Our project specifically tackles the STEM Explorations: Roller Coaster Set, which is a 546-piece model with guided labs [2]. K'Nex are used here to “boost their (children’s) understanding of the STEAM concepts associated with a real-life amusement park ride.” One experiment from the set guides the users to build a single lift coaster [16]. This experiment achieves the described objective of “determine(ing) the relationship between the mass of an object and the height above the ground.” The general understanding of energy transfer is introduced, but the observed kinematics fall short in accuracy. K'Nex are marketed related to “real-life amusement park ride(s) [2];” visually misrepresentative kinematics are detrimental to K'Nex models’ power in education.

5. Filament Extruder Assembly Components

The Creality Filament Extruder Assembly consists of 6 main components: base, lever, idler, extrusion gear, spring, and a motor [17]. This base of the mechanism is affixed to a stepper motor face, and an extrusion gear is attached to its shaft. The lever is then pinned to the upper left corner of the base such that it can pivot about this point. On the right side of the lever, a compression spring acts as a tensioner, and an idler on the right serves to guide the filament. Turning the screw that goes through the spring adjusts the amount of force applied between the idler and extrusion gear. When the stepper motor is turned on, it grabs the filament and pulls it between the two wheels. This assembly is shown in Figure 44 below.



Figure 4: Assembled and Disassembled Creality Ender 3 Extruder [17]

Initial Designs

1. Driving Mechanisms

The various methods of physically altering the trains kinematics are discussed below.

1.1 Rack and Pinion

Our initial idea was to drive the train with an on-board motor that was driving a gear. This gear would connect to a toothed rack shaped to the K'Nex track. While full control was possible, the

complex track geometry meant that toothed interactions would not fit on tight bends or twists in the track. Another issue was the requirement for the track layout to be consistent among each base K'Nex set. A visual of this assembly is shown below in Figure 55.

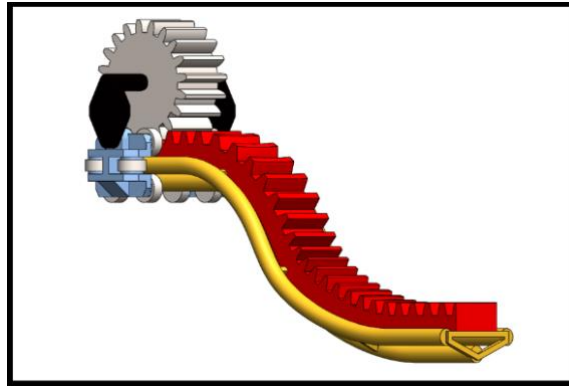


Figure 5: CAD Model of Rack and Pinion Mechanism

1.2 Friction Wheel

To account for the issues with a toothed rack, we suggested a friction wheel mechanism. With the same on-board motor but instead driving a wheel, the wheel would interface with a flat, high-friction surface. In this design, full velocity control would not be possible but bends and twists would be more easily designed. However, one drawback of this design would be that the friction wheel would still require consistent track layouts. Additionally, the freedom between the cart and surface creates the potential for the cart to not complete the track. Being limited to the frictional force tangential to the driving wheel, any variations in the normal force may contribute to a loss of control. It should be noted that the frictional constant between the wheel and surface would not change throughout the cart's motion as it is material dependent but is still a limiting factor in feasibly implementing this design.

1.3 Filament Extruder Mechanism

Based on a 3D printer's filament extrusion mechanism, this design interacts with the K'Nex track rail. In a 3D printer, the fixed component is the extrusion wheel, and the plastic filament moves. When used on the train, the rail (modeling the filament) is fixed, and the train (containing the extruder) is free to move along the rail (filament). In this mechanism, the previously passive road wheels would be driven by the on-board motor. As opposed to adding a new gear or wheel, the passive road wheels reduced the alterations to the K'Nex train. This solution works for tight bends and does not require consistent track layouts. With the interaction occurring between the track itself, the train's motion can be altered regardless of the layout. Another benefit of this design is its more direct relationship with real rollercoaster trains. Previous designs with extra wheels added components not on typical rollercoaster trains, but this design integrates the driving mechanism seamlessly.

2. Filament Extruder Wheel Assembly

The following designs (Other than the initial stepper motor test) were developed with a previously selected 12V 3000RPM Anreack DC motor [18]. This motor's form factor was modeled in SolidWorks and was intentionally placed within the cart's body.

2.1 Smaller Tooth Gear and U-Shaped Idler (Stepper Motor Test)

This gear and idler assembly was based on a preexisting 3D printer filament extruder (Creality Ender 3 [17]). After modifying the layout to work on a Nema 17 Stepper motor, the gear and idler were used to move a loose piece of straight K'Nex track; this adjusted assembly is shown below in Figure 6.

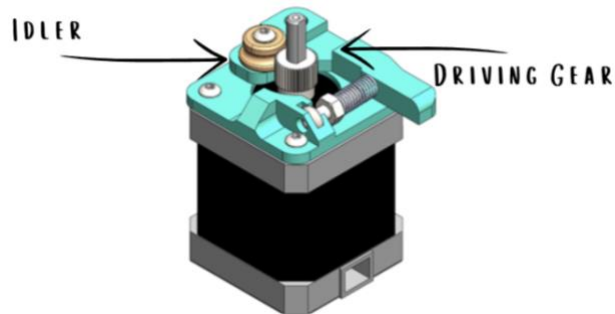


Figure 6: Filament Extruder Trial Assembly

Despite not fully integrating with the train assembly, the mechanism proved promising as it was able to translate the piece of track easily. However, upon inspection, it was determined that the depth of the u-shaped idler hole was too deep as it was interfering with the cross ties of the rail.

Additionally, the 3D printer resolution affected the small pitch of the gear and kept the gear too smooth to reliably drive the track. This solution was not constructed to be used on the train, but rather to learn about the mechanism idea and the feasibility of taking the theoretical idea to a practical application. This testing highlighted the importance of considering 3D printing tolerances.

2.2 Smooth Drive Wheel and Idler

The first on-train integration involved a two-wheel mechanism that was fully 3D printed, showing the general idea of a motor fitting over a wheel assembly to interact with a track beneath. This iteration was the first proof-of-concept but required modifications to become more effective. Most notably, it highlighted the difficulty of integrating the compression spring on the side to align the wheels properly. This assembly is shown in Figure 7.

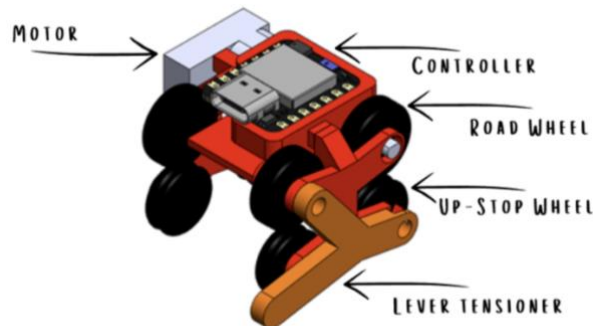


Figure 7: Model of Smooth Drive Wheel and Idler

To apply the driving mechanism to the train effectively, the side wheel supports were designed using a base sketch that is dependent on the size of the respective wheels. As a result, this model could be easily adjusted to future iterations with different assembly components.

2.3 Toothed Drive Wheel and U-Shaped Idler Wheel Assembly

This mechanism improved upon the connection between the assembly and the track by using a toothed drive wheel for increased friction. This design was printed as shown in Figure 88 below.

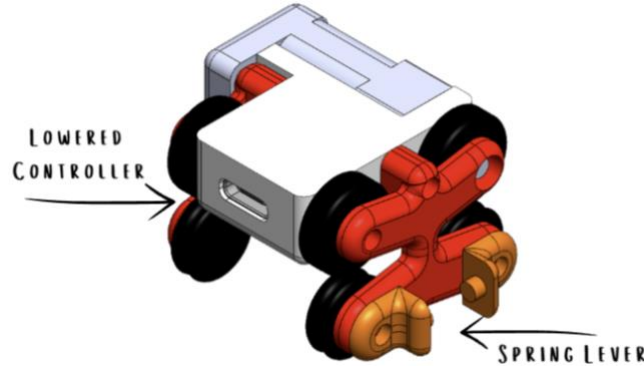


Figure 8: CAD Model Toothed Drive Wheel and U-Shaped Idler Wheel Assembly

Although no motor was available for the test, hand driven motion tests showed promise for this design choice. The curved idler end did not print well due to the thin layers at the tapered edge. This flaw was a learning moment for 3D printing resolution. Additionally, this feature was added to aid the rolling of the up-stop wheel, but upon reflection it was not effective at that goal. After removing the tapered edge and using the base up-stop wheels, this mechanism was a step forward in being able to reliably interact with the track rails. It should be noted that the tapered up-stop wheel had a second purpose to prevent train misalignment, acting as a bumper. Consequently, its removal presented this potential failure point.

2.4 Front Cluster Side-Guide Wheel Addition (With Spring)

To account for the potential train alignment issues, a side-guide wheel was added to the toothed drive wheel assembly. This design is shown in Figure 9.

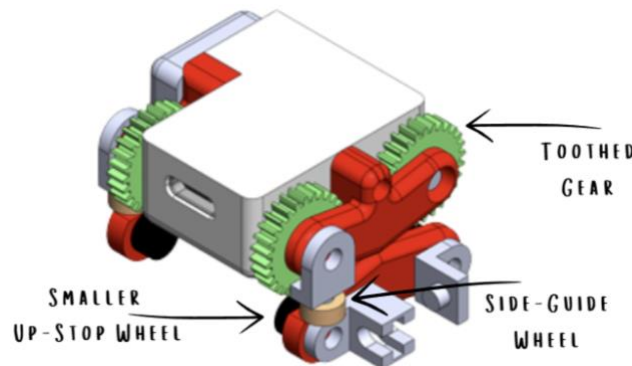


Figure 9: CAD Model of Front Cluster Side-Guide Wheel

To add the side-guide wheels, an outer support component was added to the outside of the top-front wheel connection point. This component was modeled based on the previously designed

spring holding lever. To make space for this new component and the side-guide wheel itself, the spring holding lever was reduced in profile. The side-guide wheels were not designed to be in constant contact with the rail but rather a smooth interface on curves and bends to reduce energy loss in these complex track features. The additional side-guide wheels were selected to be off-the-shelf bearings (3mm ID; 6mm OD; 2.5mm Thickness). With low internal friction and tight tolerances, these off-the-shelf models were a great choice for aligning the train without losing significant kinetic energy. Another change of the wheel assembly was moving the intended up-stop wheel outer diameter from 8mm to 6mm. After modeling the side-guide wheel as a bearing, the up-stop wheels were changed to be the same bearing.

2.5 Centered Side-Guide Wheel

After adding the side-guide wheels in CAD, issues with spacing and feasibility of assembly were noted. To accommodate these issues, current K'Nex designs were consulted, and a new side-

guide wheel integration was designed (See Figure 10 below).

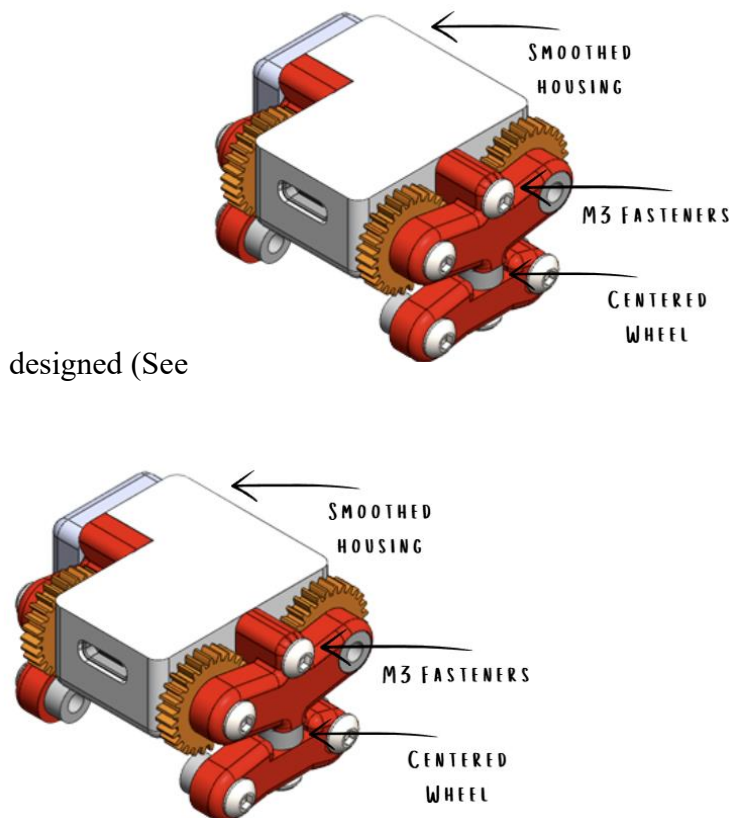


Figure 10: CAD Model of Centered Side-Guide Wheel

Adequate space for the side-guide wheel was developed by offsetting the side-guide wheel from the road and up-stop wheel pairs. With centered side-guide wheels, this new assembly allowed for increased contact between the side-guide wheels and the track. This provided the benefit of reducing the potential for train misalignment. To place the side-guide wheel here, the wheel holding component was split into top and bottom parts; the side-guide wheel was placed in between these two components and held with a dual purpose M3 screw that was also used to hold the top and bottom components together. This wheel cluster left the upstop wheels free to rotate as they were constrained by just one point. This freedom contributed to running issues in initial testing.

2.6. Crown-Structured Wheel Cluster

A dog clutch mechanism was introduced to allow the motor to be fully disengaged from the drive wheel, enabling a frictionless operating mode. This frictionless mode could provide educational benefits to show the even and uneven scaling of kinematics on the same cart and track assembly. Additional refinements were made to improve usability and automatic operation. A lever mechanism was incorporated to disengage the dog clutch, featuring chamfered geometry that enables automatic unlatching when the motor is driven, as rotational motion forces the latch outward. A spring was also added to the lever to ensure reliable re-engagement when the clutch is pulled back. Finally, a manual latch release button was integrated to provide direct user control over disengagement.

The upper and lower wheel assemblies were structurally connected using a truss-like framework, increasing rigidity and alignment stability. More specifically, the upstop wheels were constrained to run in-line with the direction of travel. This was the key improvement in the wheel cluster from the model described in Section 2.5. Material was removed from the inner face of the wheel cluster to expose the guide wheel bearing surface, simplifying access and potentially reducing weight. Additionally, the road wheel diameter was increased to improve undercarriage clearance, reducing interference with the track.

The motor mount was updated with a snap-fit faceplate to enclose and secure the electronics. This design improves accessibility while maintaining a clean and integrated housing for electrical components.

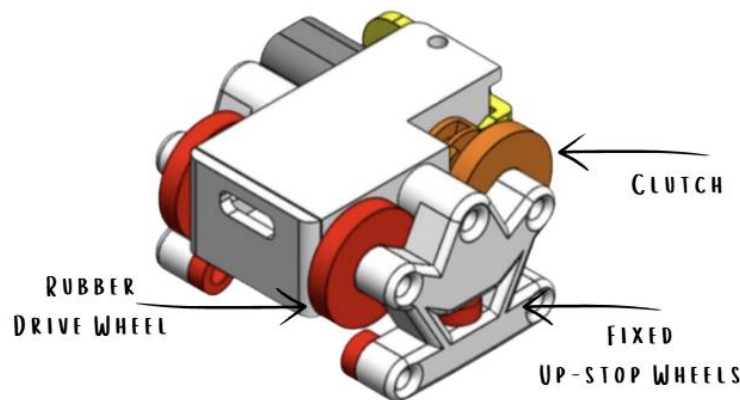


Figure 11: CAD Model of Crown-Structured Wheel Cluster

2.7. Swivel Cart

This cart most directly resembles the original K'Nex cart. By following the initially functioning cart, this design more easily integrates with K'Nex track components. The cart body was split into a front and back segment to provide a rotational degree of freedom about the axis aligned with the direction of travel. Connected by an M3 Screw, the front and back segments are designed to hold required electronic components.

This cart contains a road, guide, and up-stop wheel on all 4 corners of the cart to support smooth and consistent contact with the track rails. The driving wheel is not directly connected to the motor, but through a gear train assembly the motor's torque is applied to the driving wheel. The driving wheel is connected to a shaft that holds a 24-tooth gear on the other end. This gear interacts with another 24-tooth gear that is directly attached to the motor. The 1:1 gear ratio assembly allows the rotational motion to be transferred to the road wheel below the motor, while keeping the motor above the track.

While the increased rigidity from the crown structures wheel cluster aided in alignment on straight track segments, the cart was unable to run smoothly through complex track geometry. The swivel cart provides the rigidity of each individual wheel assembly while maintaining freedom in the cart body to accommodate hills, valleys, and turns.

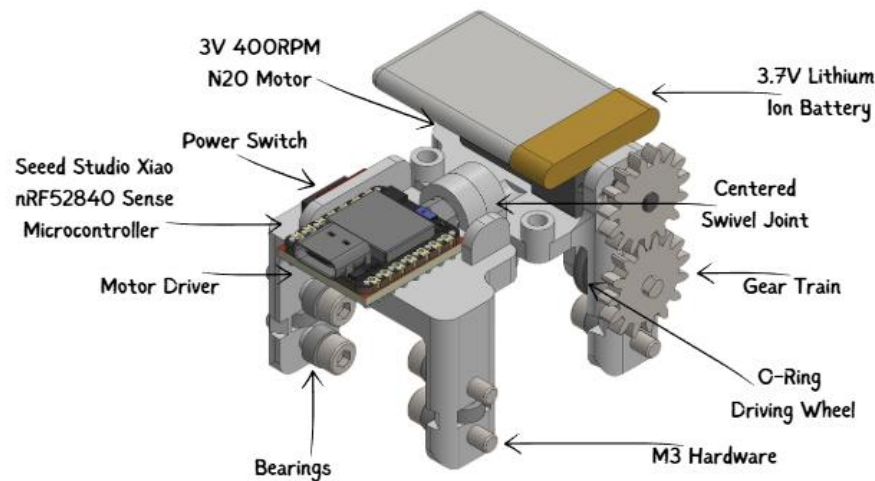


Figure 12: CAD Model of Swivel Cart

3. Top Shell Design

The following section describes the components that will rest above the driving and wheel assemblies on the train cart. At the time of writing, this includes model seats and a battery holster represented in CAD in Figure 133 below.

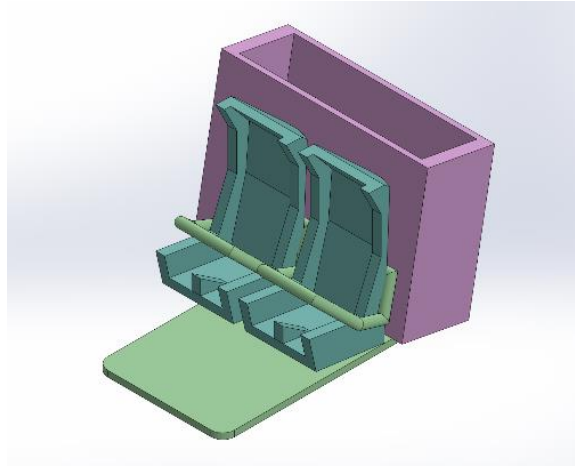


Figure 13: Initial CAD Model of Top Shell

3.1 Model Seats

Using the contour of the top of the electronics housing, initial (aesthetic) seats were designed. These seats serve no structural purpose and are planned to be 3D-Printed components once finalized. This addition to the previous designs acts to increase the visual accuracy of the modified K’Nex STEM Explorations Set. Design choices to aid in visual accuracy include the inclined seat orientation, restraint placement, and scale. The scale can be easily adjusted in SolidWorks before printing and is expected to be done near the end of the project’s completion.

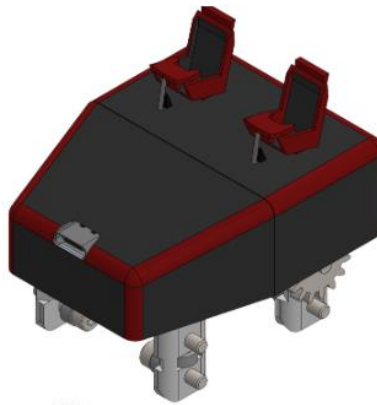


Figure 14: Revised CAD Model of Top Shell

The final wheel assembly design dramatically changed the form factor of the cart. Changes include splitting the cart into 2 sections, elongating the cart, and rearranging the electronics. These changes were met by the design shown in Figure 14 above. The new top shell is also split into 2 segments: a front shroud and a back seat section. The front shroud is designed to hold the MCU and motor driver. The back seat segment contains the buck converter and motor itself. With wires between the segments this shell hides electronics visually and protects components from potential damage. The seats on the back segment can be 3D printed separately and glued on to allow for printing of complex geometry. The seats are purely visual but contribute to the project as visual accuracy is the target method to increase educational value.

3.2 Battery Holster

3.2.1 Battery Holster Purpose

The initial battery holster is integrated into the model seats' support system. The general design includes a surrounding 3D Printed shell with a slot on top for battery placement. Wiring details to connect the battery to the motor are not flushed out in this design. This 3D draft is not ready for integration but is helpful for holding the battery once the battery's scale is selected.

3.2.2 Battery Holster Final Decision

While functional for previous iterations, the current electronics layout does not require a battery holster. By securing the battery on top of the motor and expanding the shell's dimension, the electronics can be completely enclosed. Future iterations may reintroduce this concept, but as of the end of phase 1, is no longer needed.

4. Modelling

4.1 MATLAB Code

The modelling of the system will be done in MATLAB but will utilize different software programs to turn the track into a usable system of coordinates that can calculate the speeds and motion of the system. To start, we have the image of a 2D version of the track, as this will be able to be converted to a spline in SolidWorks. By uploading this image in SolidWorks, we can layer a sketch over the image and create a spline based on certain predetermined size constraints of the coaster. This action is shown in Figure 155.

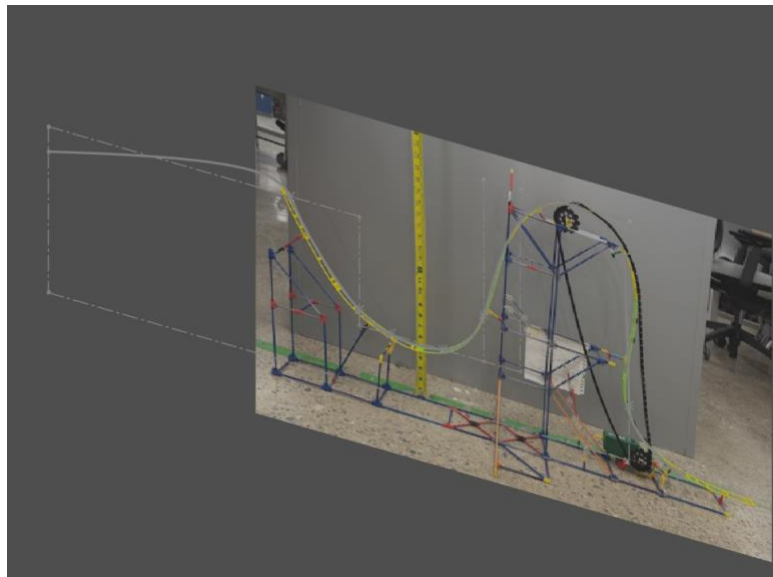


Figure 15: SolidWorks Sketch of 2D Track Image

Once this spline is made, we can export it as a DXF file which saves the size of the spline, but removes unnecessary parts of the file like the image (See Figure 166 below), and we can upload it into AutoCAD, to find the XY Coordinates. Once the file is uploaded, we will split it into segments using the **divide** command. After dividing, we can use the **DataExtraction** command to be able to extract the points of each of the divided segments.

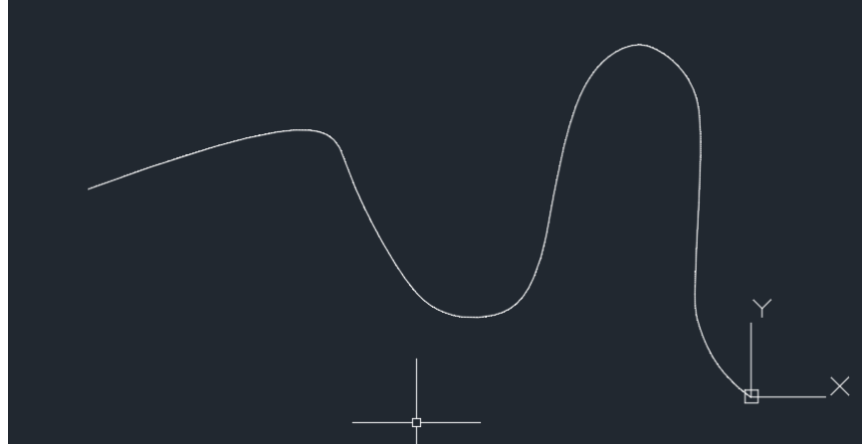


Figure 16: Resulting Spline of the SolidWorks in AutoCAD

Once we have the values of all the points, we can import them as a CSV file into MATLAB to start simulating a cart going through the track. These points can be visualized using the plot command in MATLAB to ensure the CSV points from AutoCAD are consistent; the resulting plot is shared in Figure 177.

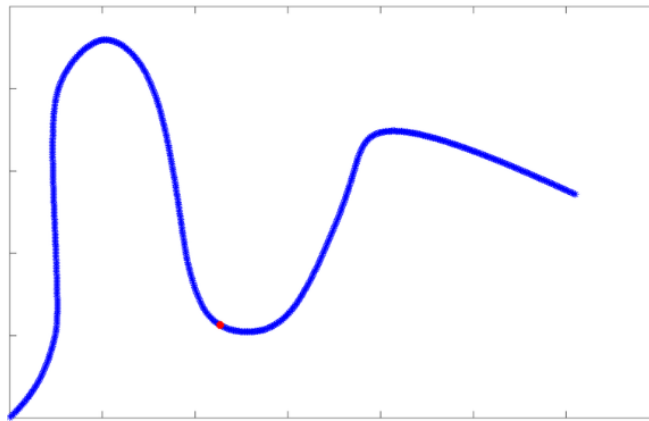


Figure 17: MATLAB Plot of 2D Spline

Since this is a gravity driven coaster, we start the simulation at the top of the highest point and calculate the maximum potential energy. With the maximum potential energy, we increment the train forward a single point to calculate the new potential energy value. With this lower potential energy, we can then calculate the kinetic energy and speed at that point. Looping this concept, we calculate the kinetic energy and potential energy at the spots throughout the coaster.

4.2 MATLAB Friction

Current roller coaster simulation software uses a friction factor that relates the distance travelled to the amount of dissipated energy - in terms of height. FVD++, one of the software, uses friction coefficients ranging from 0.024-0.03. A value of 0.025 would mean that after traveling 100 meters of track, frictional losses would dissipate 2.5 meters of potential energy. During the iterative calculations of velocity along the track which use the change in height to describe the change in potential energy, a term can be added to represent the frictional loss between the two points based on this factor and distance between these points.

To determine time and other values, we can make a few assumptions that will help determine the values. For example, we can assume that the velocity vector is tangential to the track, and the next point is a straight distance. This means that by calculating the distance and dividing by the velocity, we find the time from one point to the next. The resultant time series created compares the velocity, displacement, angle, potential and kinetic energy. Using the time series, we can linearly interpolate any location on the track to find the theoretical speed of the cart. With a decreased step size, the resolution increases, and the assumptions of tangential velocity in each iterative calculation increases in validity.

4.3 Scale Factor

The scale factor is an input in the function above since it helps determine the speeds of the train on specific coasters, but this project is focused on determining how the train speed will change relative to the size of the coaster. So, as part of the spline above we take the highest point and divide the size of the coaster, height, and length, by that value to get a 'unit' coaster. From there we can multiply the length and height by the theoretical height of the necessary coaster. With this scale factor we can calculate the new speeds of a coaster of this scaled height.

With this new function, we can simulate this path at many sizes and apply the descaling factors to properly understand the speeds they should move to illustrate the larger coaster on a smaller model.

While the scale factor calculations are based on a simplified model (See Research Section 2.1), further details are not necessary to create visual accuracy. Friction has been researched and can be implemented but is not yet necessary at the state of this cart's design.

5. Control Mechanisms

The methods to control the cart through the motions are split into 3 categories. The first is open loop, the second is closed loop, then the last is bridge, which utilizes multiple sensors to control the cart.

5.1 Open Loop System

While less robust in its ability to control the train across any track layout, by knowing the desired motion of the train on the K'Nex STEM Explorations Set layout, we can preprogram the desired motor outputs. This method is the simplest in engineering rigor and the most easily achievable. The limited adjustments available require that the track is consistently built for each use, but due to its simplistic integration increased testing may be possible. While true kinematic accuracy is desired, the educational aspect of the train's alteration may not require perfect kinematic accuracy. This tolerance in the train's motion to still achieve the educational and visual changes may leave this solution as the easiest solution, while still being impactful.

5.2 Closed Loop System

To abstract the track layout from the controlling of the train, we can develop a program to control the train's motion based on its current velocity and angle of travel. The train's onboard 6-axis IMU provides current information of the train's orientation and acceleration (which can be used to estimate the velocity) at any point in its motion. It should be noted that this velocity estimation loses accuracy with time. With this information and proper manipulation of engineering dynamics theory, a closed loop system that can control the train's motion in real time is possible. This idea, if successful, would be much more efficient and adaptable to any track layout. The

cost of these benefits is seen in the difficulty of integration. Additionally, the ability to integrate this system on varied track layouts may be limited by physical driving factors as described above; with limited integration from other factors, this benefit would be a moot point.

5.3 Bridged System

A bridged system utilizes sensors along the track to register the cart's location. Through including an emitter on the cart - hall effector or NFC antenna - and sensor - a signal, magnet, or NFC chip - on the track, we will be able to understand the location of the cart along the track. This will ensure that the IMU signals are reading correctly, and we can confirm that the values that are entering the Closed loop system are correct. Additionally, by using the space of time between registered locations, the velocity can be interpolated to aid in the location and velocity mapping of the cart.

5.4 Control Mechanism Implementation

Once the MATLAB Script has a map of the points along the roller coaster (described in Section 4.1 above), we can use the code to calculate a couple different parts. First is using the graph to make a video that illustrates the uneven scaling, second is using these scripts to find the Velocities and convert them to the necessary PWMs. And finally, the with the conversions of the PWMs from velocities, determine the motors and their relative scales that work well for the range of speeds. The final script is described in a later section as it refers to motor selection.

First is the script that makes the video. The initial design of this script takes the distance in between the points, and from the known velocity from energy laws, we can calculate the Δt between each of these points. With this value we have the known position through which the cart moves, and although it is steps through which the cart moves, we can break down the locations and the time and linearly interpolate to find the location of the cart such that the position is shown every frame. With this we can find the location at set time intervals, instead of it being calculated based off intervals of distance. We then set the start location at the highest point of the coaster to show the gravity driven nature of the coaster.

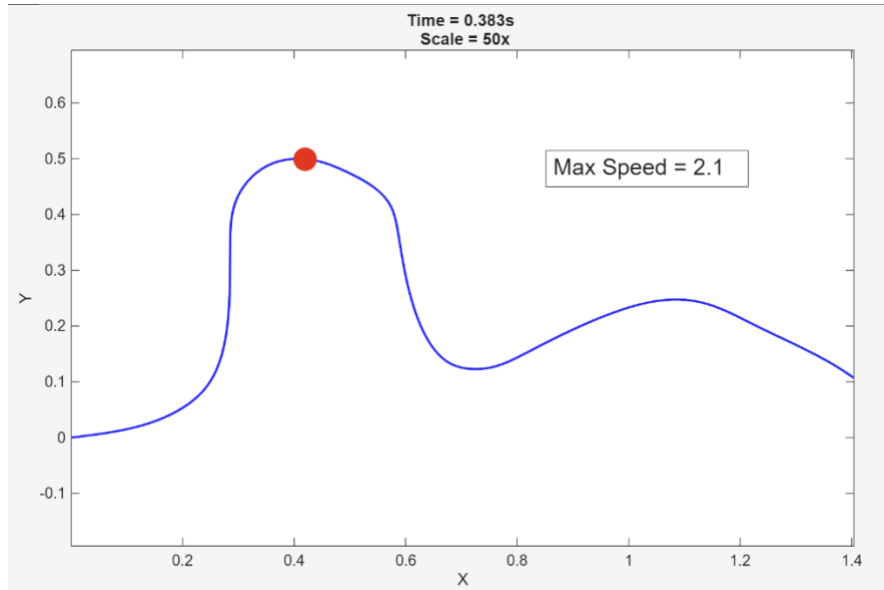


Figure 18: X and Y Values of the Track Spline Plotted with Maximum Speed

The second script then takes the initial values from the velocity, time, and position arrays and linearly interpolate so that we have a recording at a constant time interval of 1ms. With this data and the motor characteristics we can calculate the necessary PWM signals for each velocity along the coaster. This script takes the input of the positions and exports a csv file that has a list of the PWM and the delta time for which to hold the speeds. These values are then recorded in a CSV file with columns, index, PWM value, and PWM time. Using AI and the microcontroller datasheet, a web app was created to facilitate Bluetooth connection to the microcontroller, allowing seamless transmission of the PWM values to the onboard motor [19]. On the firmware side, the AI was used to help formulate logic to receive the CSV transmission from the web app in accordance with the documentation from the microcontroller's manufacturer [19].

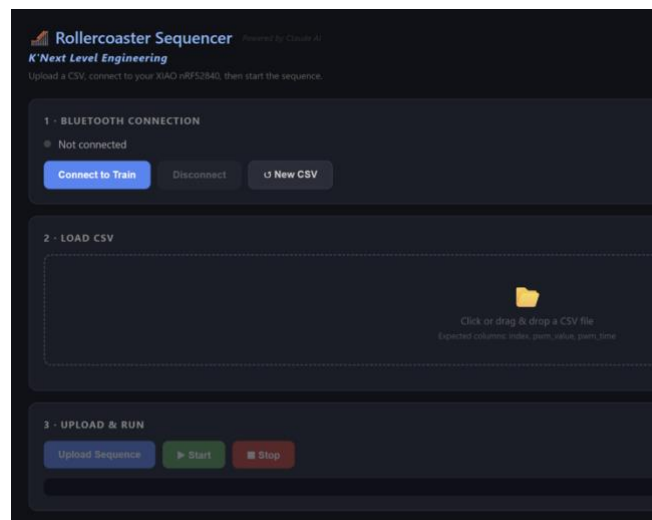


Figure 19: Web App for CSV Transmission

Decision Criteria

1. Driving Mechanisms

The driving mechanism was selected in the first week of Capstone 2 to be the filament extruder mechanism. This option was chosen based on its ability to run and be controlled on various track layout components (hills, valley, turns, etc.) as well as its likeness to true roller coaster wheel assemblies and expected ease of integration. Although other ideas may be explored if dramatic unforeseen issues arise, the running model for Capstone 2 has and will be the filament extruder mechanism.

1.1 Filament Extruder Wheel Assembly

The filament extruder wheel assembly will be assessed based on its reliability of motion control and ease of construction.

More design criteria, other than slippage, do not require structured testing, as functional operation will confirm or deny whether the assembly acts as desired. However, the reliability of motion control will be evaluated rigorously using 3 different track layouts of varied complexity (See Track Design Section for more). More specifically, the cart will be run with predetermined changes to the expected gravity-driven motion. With repeated runs the reliability of a model can be analyzed. If the driving mechanism cannot engage properly to control the cart's motion as desired, test with the same control system input (predetermined changes from gravity-driven motion) will show different outputs in the cart's motion. Any variation in motion will be noted and discussed for significance to determine if the cart meets design requirements.

Ease of Construction is a qualitative trait that can be assessed based on team member reflections during the initial construction process. If time permits, testing assembly on less familiar users may provide deeper insight into the true ease of construction. While this trait is desirable, the cart's function is paramount in meeting the problem statement.

1.2 Top Shell Design

The seat design will be chosen based on two qualitative criteria: feasibility of construction and aesthetic appeal. The feasibility of construction will be assessed by 3D printing the designs. A proper design will print similar to its CAD model; it also can be removed from the printing bed and supports without degradation of the part. The aesthetic accuracy and quality will be assessed with feedback from: team members, Capstone faculty, test users, and K'Nex hobbyists (if time allows).

The battery holster design will be chosen by printing and assessing its ability to hold the intended battery. A successful holster will allow for simple wire management. Additionally, holster location and/or orientation will be considered in its effect of weight distribution of the train.

2. Control Mechanisms

The control system will be analyzed through physical testing and through creating a digital twin to understand the limits of the control system. Through the creation of a digital twin, we will be able to see the theoretical upper limits regarding how the controls can react to changing outputs (speed up or slow down).

This experimental baseline forms a quantitative foundation for comparing design alternatives and validating whether the implemented control system achieves the intended educational and kinematic objectives.

Final Design

1. Cart Design

The recommended design (See Figure 21 below) uses a motor-driven filament extruder wheel assembly integrated into the coaster cart, where an O-Ring rubber drive wheel presses against the track rail to actively control the cart's speed as it travels along the track. The system uses a small DC motor (N20 3V 400 RPM), a gear train (Both: 1mm Pitch - 15 teeth), a 3D-printed driving wheel component with an O-Ring rim (Back Left), and bearings to create a compact wheel cluster that both guides the cart and provides controlled propulsion. The central swivel joint provides rotational freedom along the direction of travel. This aids in traversing complex track geometry without interference. The wheel cluster support legs also have chamfers along the guide wheels to provide clearance on turns. The gear train allows for the motor to be held above the track and aids in electronics layout. The front half of the cart holds the Seeed Studio Xiao nRF52840 Sense microcontroller, Motor Driver and Power Switch, while the back half holds the motor and battery. These electronics are presented in the wire diagram (Figure 2222) below.



Figure 21: Final Cart Design

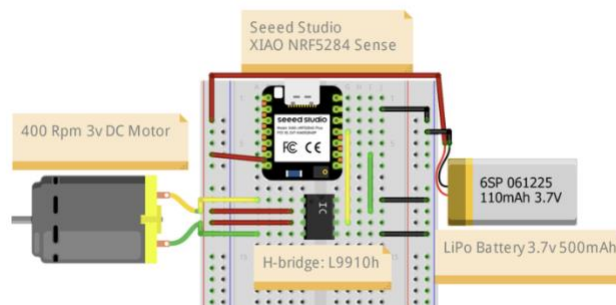


Figure 22: Wire Diagram of Completed Assembly

This cart model produces sufficient contact between the wheel assembly and the track to ensure smooth motion. This proper contact also aids in ability to directly relate the motor's action to the cart's motion. The chosen motor provides sufficient torque to drive the cart while fitting within the small cart constraints. The central rotational freedom about the direction of travel allows the cart to run without interference from the track.

Additional detailed images of the completed assembly can be found in Appendix A.

2. Controls Design

Analysis & Testing

1. Track Construction

1.1 Prototype Track

Integrated testing has not been conducted at this time, but efforts to create the necessary testing materials have been completed. This investigation of materials includes developing three track layouts with varied complexity; each track has an additional dimension.

As described in the K'Nex Modifications section of the Literature Review section, each track layout was completed using the lessons learned from various K'Nex hobbyists. Most specifically, tutorials from the K'Nex Enthusiast were consulted to develop a 'spine' along the track pieces. This spine allows the track to be easily connected to supports; This better connection provides two benefits; the supports reduce track movement and energy loss when properly connected to the track; increased connection points allow for more freedom in designing the track layout.

The 1D track is a singular piece of straight track and will serve for quick and informal tests during the design process seen in Figure 23 below. The 2D track, consisting of an initial drop followed by a short hill shown in Figure 24 below, will serve as the proof-of-concept layout. 3D track, as seen in Figure 25 below, is the goal operational layout. This track is the STEM Explorations layout and is directly applicable to the base K'Nex set and learning experience described in the problem statement. Modifications to reduce the 3D track's movement during operation are expected to be made on an as needed basis during 3D testing; initial modifications will be done once the cart assembly is ready for 3D testing and will be based on lessons from the K'Nex modifications used on the 1D and 2D track construction.



Figure 23: 1D Track Layout

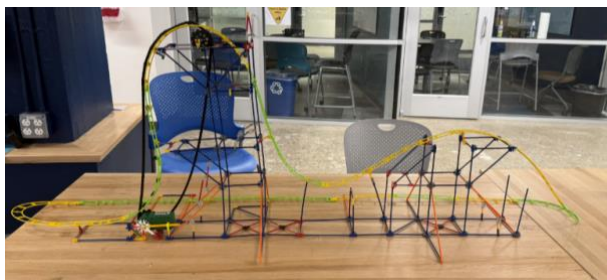


Figure 24: 2D Track Layout



Figure 25: 3D (K'Nex STEM Explorations) Track Layout

1.2 Final Track

Two final test tracks were developed to evaluate the performance of the cart system under distinct motion conditions. These tracks were designed to represent simplified two-dimensional projections of potential real-world motion, with the goal of demonstrating that successful operation in both cases would indicate feasibility for extension into a full three-dimensional system.

The first configuration, referred to as the XY track seen below in Figure 26, consisted of a linear path with varying elevation, including hills and valleys. This track was designed to test the cart's ability to handle changes in gravitational potential energy and to evaluate performance under continuously changing slopes. The varying height profile introduced dynamic forces along the direction of motion, making it an effective representation of non-uniform terrain.



Figure 26: XY Track Model

The second configuration, referred to as the XZ track seen below in Figure 27, was a flat circular loop. This track was designed to isolate lateral and centripetal motion while maintaining constant elevation. By removing vertical variation, the XZ track allowed for evaluation of sustained motion through turning, highlighting stability and control.

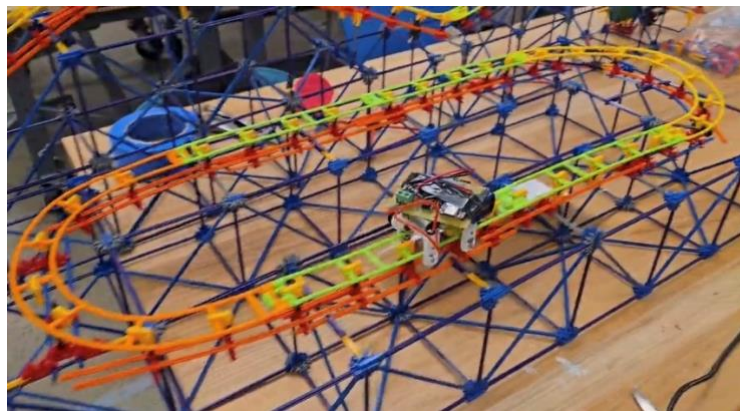


Figure 27: XZ Track Model

Together, these two tracks were intended to demonstrate that if the cart could operate successfully under both varying elevation (XY) and curvature (XZ), then it would be theoretically capable of functioning on a combined three-dimensional track system incorporating both effects simultaneously.

Both tracks were constructed with a strong emphasis on structural reinforcement to minimize deformation during operation. Given the scale and complexity of the system, multiple reinforcement strategies were implemented, drawing from established building techniques used by experienced K'Nex designers. These strategies included increased support density at high-

stress points, cross-bracing along long spans, and redundant structural pathways to distribute load more evenly across the framework. This ensured that track deformation would not significantly influence experimental results.

2. Tracker Analysis

2.1 Tracker Familiarization and Initial Software Testing

Initial analysis efforts focused on learning and validating the use of Tracker video analysis software as a tool for quantifying coaster motion. Because future design decisions will rely on accurate measurement of velocity and acceleration, it was necessary to first confirm that Tracker could reliably capture and process motion data for the K'Nex coaster.

Preliminary video recordings of the 2D test track were imported into Tracker and analyzed to gain familiarity with calibration, scaling, coordinate system alignment, and point-mass tracking. During this process, several key lessons were learned regarding best practices for reliable motion tracking. These included the importance of consistent camera positioning perpendicular to the plane of motion, maintaining fixed lighting conditions to improve point visibility, and ensuring a clearly defined scale reference within the video frame.

The biggest pain point with the preliminary test was that Tracker relied on a set “template” to follow throughout the video. During initial use of Tracker, it became clear that successful motion tracking required a consistent visual template for the software to follow throughout the video. Tracker relies on a pre-defined, reusable template of an object’s visual features to detect and track that object across frames. In early trials, the template was defined using the yellow cart body and its diagonal black stripe. However, as the cart moved through the track, its orientation relative to the camera changed significantly. Because the diagonal stripe rotated with the cart, the visual appearance no longer matched the original template, causing Tracker to lose the object once it passed the initial lift hill and began descending and changing angle. This resulted in incomplete tracking data and highlighted the importance of maintaining a consistent visual profile for the tracked object.

To address this issue, a high-contrast tracking marker was added to the cart. A piece of paper colored bright red was taped to the side of the cart, as seen in Figure 28 below, facing the camera to create a clear and consistent visual feature independent of cart orientation. This modification significantly improved tracking reliability by providing a uniform template that remained recognizable across all frames of motion.



Figure 28: Initial Tracker Test with Red Marker on Cart

Tracker exported time, horizontal position, vertical position, and their respective velocity and acceleration components. This informed the data-processing procedure used in subsequent testing. This early familiarization trial ensured that we could reliably capture repeatable motion data and establish a consistent workflow for future experimental testing.

2.2 Experimental Procedure for Baseline Cart Testing

Following familiarization with Tracker, we developed a formal procedure to characterize the baseline motion of the unmodified coaster cart. Establishing this baseline is critical for future comparison with motor-controlled prototypes, as the primary project goal is to intentionally modify coaster kinematics.

A simplified 2D track consisting of an initial drop and subsequent incline was constructed using K'Nex components. The coaster was recorded using a fixed camera positioned perpendicular to the track plane to minimize perspective distortion. A yard stick was included in the frame to allow spatial calibration within Tracker.

For each trial, the cart was released from the same starting position without external forcing beyond gravity. Three separate recordings were collected to evaluate repeatability and consistency of motion. Each video was imported into Tracker, calibrated using the known length reference, and analyzed using a point-mass tracking method focused on a consistent point on the front of the cart.

Tracker exported time, horizontal position, vertical position, and corresponding velocity and acceleration components for each frame. The data was then processed in Excel to convert the 2D motion into 1D motion along the track. The magnitude of velocity was calculated from horizontal and vertical components, and incremental changes in position were used to compute cumulative arc length along the track. Tangential acceleration was determined by differentiating velocity with respect to time using finite difference methods. This procedure provided physically meaningful kinematic quantities aligned with the direction of motion.

2.3 Initial Test Results and Data Analysis

Following the formal procedure, baseline testing of the unmodified cart produced consistent and repeatable motion across three trials. Processed Tracker data was used to generate plots characterizing the cart's position, velocity, and acceleration along the track. These plots establish a quantitative baseline for gravity-driven coaster motion and will serve as a control case for comparison with future motor-controlled prototypes. To clearly distinguish between trials, each dataset was assigned a different color on the graphs while maintaining consistent axis scales and units. Overlaying the three trials on each plot allows for visual comparison of repeatability and consistency in the coaster's motion, demonstrating that the measured kinematic behavior is reliable across multiple runs and suitable for establishing a baseline for future controlled-motion testing.

The cart's position in the X and Y directions was obtained using Tracker software and exported for analysis. This data was then converted into a 1D coordinate system along the track to better represent the physics of motion along a curved path. Arc length was calculated from the incremental changes in x and y position, allowing velocity and tangential acceleration to be evaluated along the track rather than in independent Cartesian directions.

The height of the cart was plotted as a function of arc length to generate a profile of the coaster track. This plot defines the geometry of the simplified 2D test track and serves as the foundation for all subsequent dynamics analysis (See Figure 29). The height profile confirms a clear initial drop followed by a valley and a subsequent rise, consistent with the intended test configuration. Establishing this geometry allows theoretical predictions of velocity and acceleration to be made based on gravitational effects along the track.

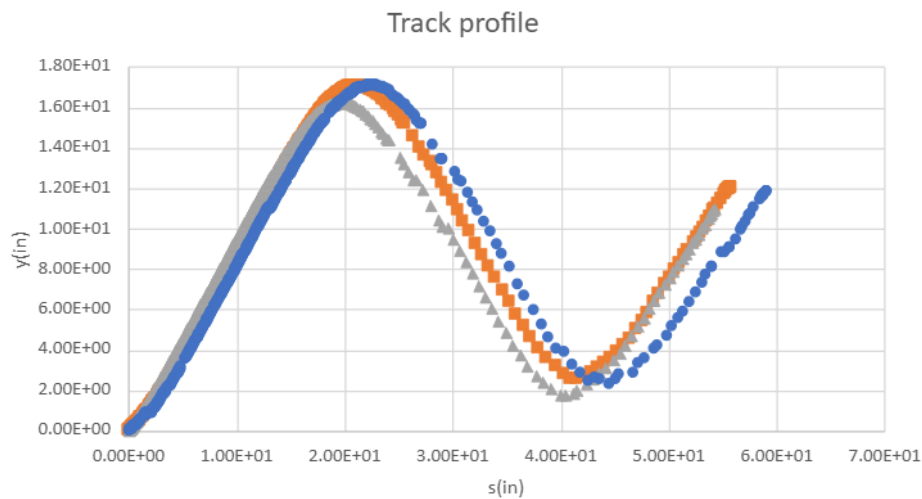


Figure 29: 2D Plot of Track Layout from Tracker Analysis (3 Runs)

The magnitude of the cart's velocity was computed from the tracked horizontal and vertical velocity components. A plot of velocity versus time shows smooth acceleration down the initial drop, reaching a peak velocity at the lowest point of the track, followed by deceleration as the cart climbs the subsequent hill. This plot is shared in Figure 30 below. This behavior confirms that the motion of the unmodified cart is primarily driven by gravity, with energy losses due to friction and rolling resistance. The velocity curve provides a clear baseline for the natural motion

of the system and will be used as a reference when evaluating the effectiveness of future motor-controlled modifications.

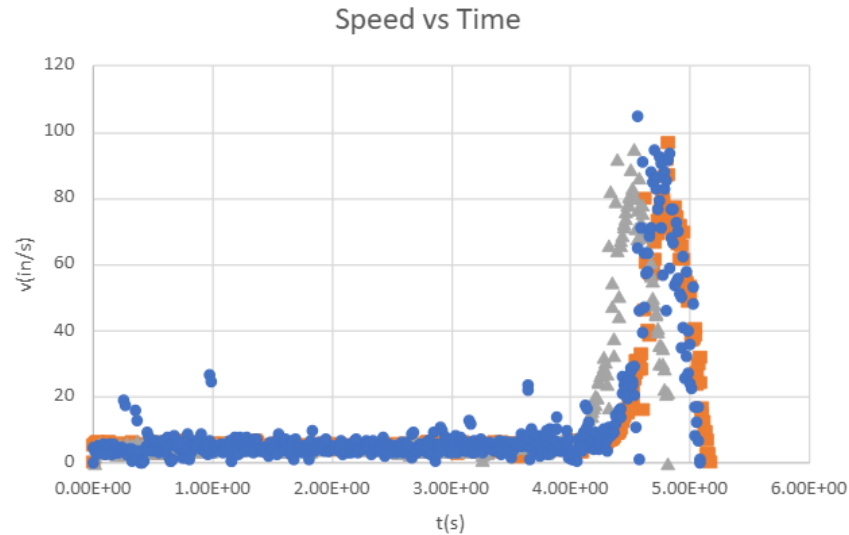


Figure 30: Speed of Cart vs Time (3 Runs)

A plot of velocity versus height was also generated to examine the motion from an energy perspective (See Figure 31 below). The overall trend shows increasing velocity as height decreases, consistent with conservation of mechanical energy. Deviations from the idealized energy relationship indicate the presence of non-conservative forces such as rolling resistance and contact friction within the K’Nex system. This plot confirms that the baseline system behaves in accordance with expected physical principles while also quantifying the magnitude of energy losses.

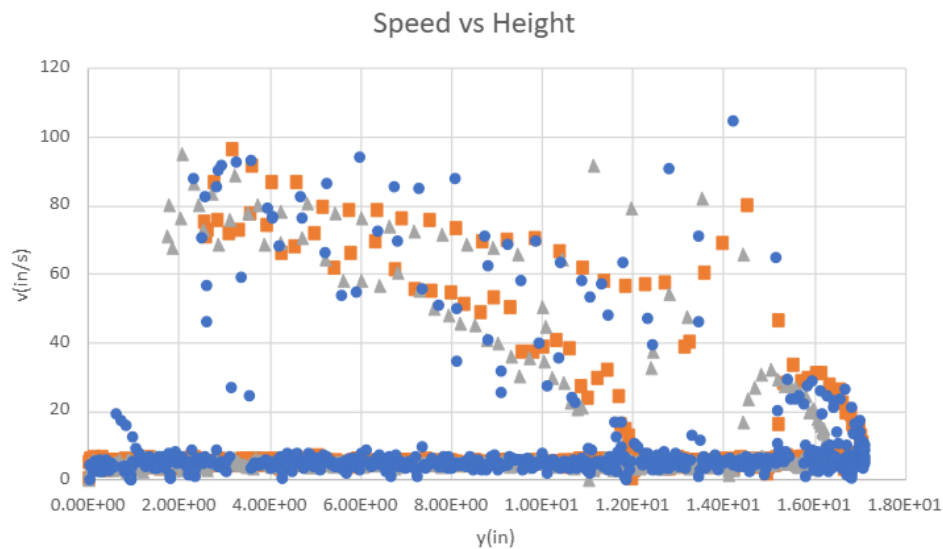


Figure 31: Speed of Cart vs Height (3 Runs)

Tangential acceleration was calculated from the rate of change of velocity with respect to time and plotted against arc length. This representation of acceleration along the track provides a more physically meaningful description of the motion than separate horizontal and vertical acceleration components and is plotted in 32 below. Regions of high positive acceleration correspond to steeper downward sections of the track, while negative acceleration appears as the cart climbs the subsequent hill. The overall trend matches the expected influence of gravitational acceleration projected along the slope of the track, with consistent reductions due to frictional effects.

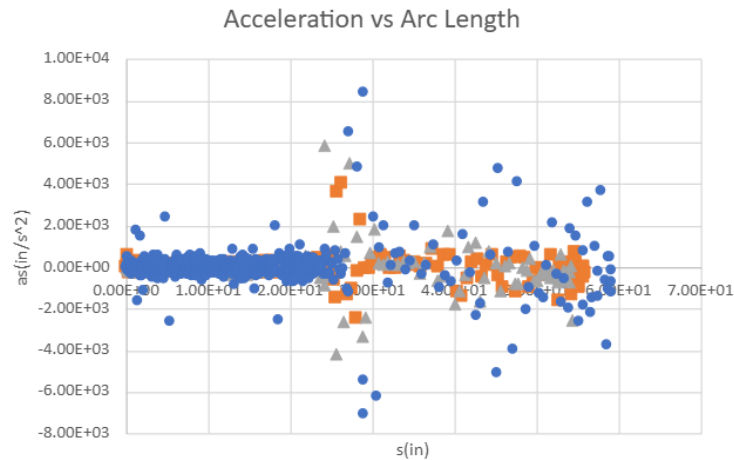


Figure 32: Acceleration of Cart along the 2D Track

Together, these analyses establish a baseline characterization of the coaster's natural motion. The tracked data confirms that the system behaves as a gravity-driven coaster with consistent and repeatable energy losses. Key metrics such as peak velocity, time to reach the lowest point of the track, and deceleration during ascent define the uncontrolled motion of the cart. These results will serve as the reference condition for future testing with the motor-controlled system, enabling direct comparison between passive and actively modified kinematics. Establishing this baseline is essential for evaluating the effectiveness of the proposed control mechanism and for supporting the project's educational objective of demonstrating kinematic principles through measurable physical motion.

2.4 Final Analysis of unmodified and motor-controlled cart on 2D track

After finalizing the cart design, system testing was conducted to evaluate the performance of both the unmodified and motor-controlled configurations on the completed test track seen in Figure 33 below. This phase focuses on comparing passive, gravity-driven motion to actively controlled motion and quantifying the differences using tracker-based data analysis.



Figure 33: Tracker Analysis on Test Track

The unmodified cart operates without active control, relying entirely on gravity and track conditions to drive motion. As a result, its speed varies significantly depending on slope changes, friction, and minor surface inconsistencies. In contrast, the motor-controlled system actively regulates motion by maintaining a target speed, reducing the influence of gravity as the primary driver of motion and allowing for a more controlled evaluation of system performance.

As shown in Figure 34, the velocity-time graph for the unmodified cart demonstrates significant variability throughout the run. The velocity is highly inconsistent, with frequent fluctuations caused by gravitational acceleration and changes in friction along the track. These irregularities indicate that the system is not actively regulating motion and instead responds passively to external forces.

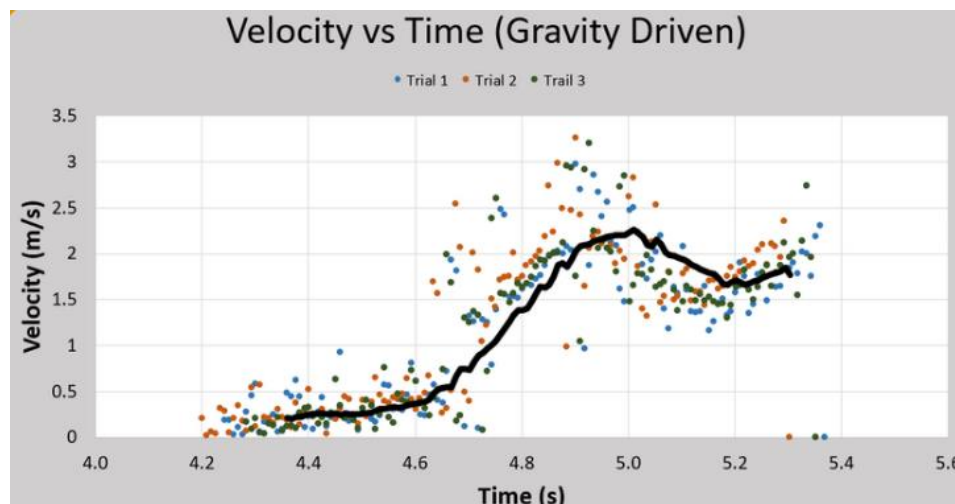


Figure 34: Velocity vs. time for the unmodified cart on the final test track

In comparison, Figure 35 shows the velocity-time graph for the motor-controlled cart. The data exhibits a much smoother and more stable velocity profile, with reduced variation over time. While small fluctuations are still present due to mechanical limitations, motor response delay,

and tracker noise, the overall trend remains consistent. This indicates that the motor system is successfully compensating for external disturbances and maintaining controlled motion.

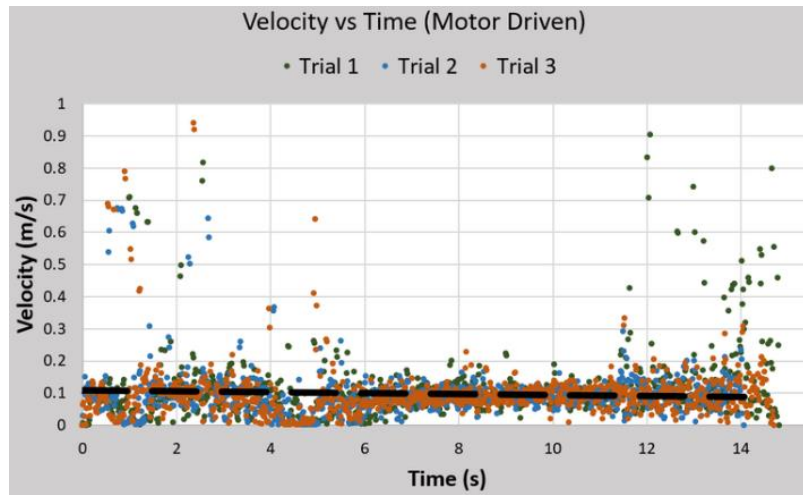


Figure 35: Velocity vs. time for the motor-controlled cart on the final test track

A direct comparison between the two datasets confirms a clear improvement in motion stability when using motor control. The unmodified cart shows uncontrolled, gravity-dominated behavior, whereas the motor-controlled cart demonstrates regulated and consistent motion. This difference is especially evident when comparing corresponding regions of the track, where the unmodified cart experiences sharp changes in velocity while the motor-controlled cart maintains a relatively uniform profile.

Overall, the tracker analysis confirms that the motor-controlled system improves motion consistency and reduces variability in cart speed. These results align with qualitative observations from testing and validate the effectiveness of the final design.

3. MATLAB Integration

3.1 Calibration

The calibration step will be used to properly map the coaster to integrate the scaling and math all into one step. We plan to take the IMU values and create a displacement plot to back calculate the necessary velocities of the cart moving through a scaled track. This will entail being able to take the IMU values and sending them to MATLAB to calculate the displacement properly. The testing track layouts will be the 3 previously described tracks with increasing complexity and dimensions (See Track Construction above).

3.2 Control System

Following calibration of the IMU and confirmation of its ability to map the track in varying dimensions, we will use this mapped data to control the cart's motion accordingly. This control system will be tested on the same 3 track layouts. Following similar theory for the calibration

steps, scaling complexity will allow for multiple benchmarks of success in the control system algorithm and implementation.

4. Motor Testing (Centered Side Guide Wheel)

4.1 Test Overview

The latest filament extruder wheel assembly (Centered Side-Guide Wheel) was equipped with the 12V N20 3000 RPM motor and was run in two configurations. In the first configuration, the cart was held in place, whereas in the second configuration the track was held in place. Testing both configurations allows for direct analysis of both torque and friction as the cart is much heavier than the track. Images of this testing set-up are shown in Figure 36 and 37 below.

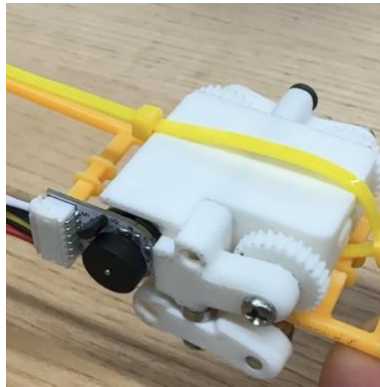


Figure 36: Testing of Centered Side-Guide Wheel Assembly (Top View)

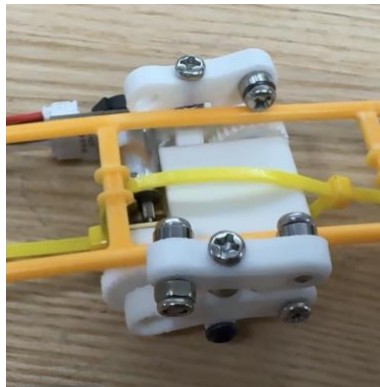


Figure 37: Testing of Centered Side-Guide Wheel Assembly (Bottom View)

4.2 Test Results

Testing revealed that the motor was barely able to move the cart along the track, slipping or stalling at any point where there was the slightest opposition to its motion. Analyzing the test setup, there were a few points that could easily be tweaked to improve the performance of similar motors with the current configuration:

1. Motor – Swap zip-tie for mounting hardware (Cart – Track Clearance)
2. Bearings - Spacers between bearings and walls (Positioning and Binding Prevention)
3. CAD Updates – Remove the ability to rotate about the screw from the wheel assembly
4. Wheels – Swap 3D printed extrusion gear for rubberized wheel (Increased Friction)

5. Spring Tensioner – Add the spring tensioner to the latest iteration (Increased Friction)

5. Motor Characterization and Selection

The following procedure was used to characterize the motors of varying RPM ratings. First the motor was wired with the circuit shown below. With a code that steps up the PWM value over a range of 20 button presses, where 1st is 25pwm, 20th is 255pwm, and the 21st button presses reset the PWM to zero. Through this we can understand the correlation between the PWM signal and the angular velocity at the output of the motor. Once this correlation is found, we can then turn the angular velocity into linear velocity based on the geometry of the wheel and the gear ratio, and in return create a regression curve to determine the PWM signal based off the necessary velocities. This testing was done at 1200rpm and 600rpm at 9v and 60rpm, 150rpm, and 300rpm motors at 3v.

A MATLAB script takes the found PWM values (for the expected track geometry), the motor characterization, and estimated motor characterizations of other motors, to create curves for a plot that illustrate the usable range of scales that each motor can function at. This helps determine the maximum speed of the motor as well as the gear characteristics and the drive wheel size. The resulting plot below shows the 400 RPM motor with the largest usable range and thus was chosen for the final design.

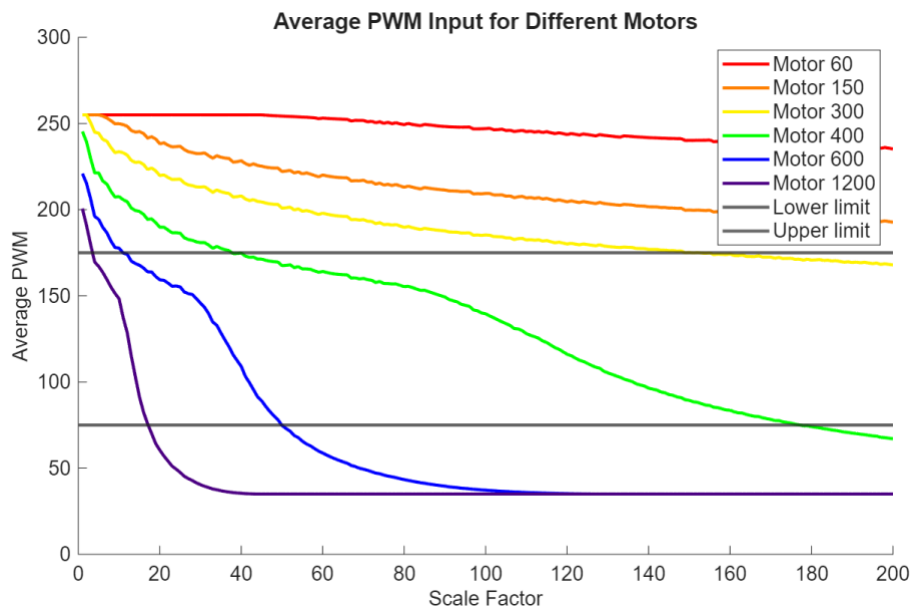


Figure 38: Average PWM Values for Different Motors

Summary and Next Steps

The current cart demonstrates feasibility for integration with the K’Nex STEM Exploration Set, providing direct application for intended users. Improvements of the cart and future expansions of the project are summarized below.

Using a different motor than originally selected has caused inconsistent torque control and runs at a slower speed than intended requiring further iterations to achieve uniform kinematic scaling; the current N20 400RPM motor does not output sufficient torque to reduce slipping in a scaled kinematic sequence. Additionally, the cart layout is sufficient, but more iterations of minor dimensioning changes would be beneficial in track interaction. More simply put, the cart does run along the micro K'Nex track, but adjusting dimensions would reduce friction issues on more complex track geometry.

Looking forward, adding a closed-loop control system would allow the model to adapt to other track layouts. Once functionality is improved, the new cart can be tested on the original goal track layout. This is the STEM Exploration Set model shown in Figure 23 above. This will apply the functioning cart to the educational field. Final improvements can come in the user experience. This may include an experiment guide outlining cart assembly, track integration, supporting processes, and the scientific purpose of the experiment to aid in the learning interaction with the cart model.

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Appendix A

Final Cart Design Images

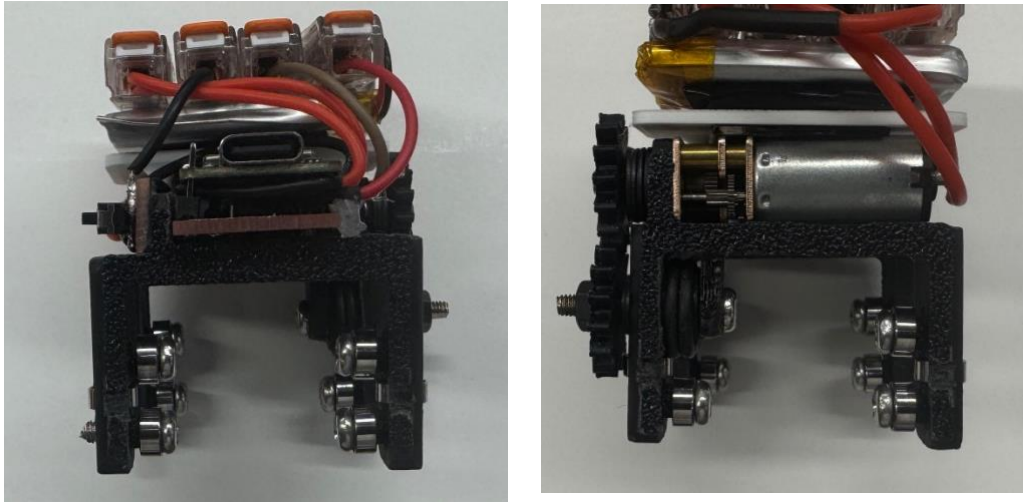


Figure A.1: Front and Back

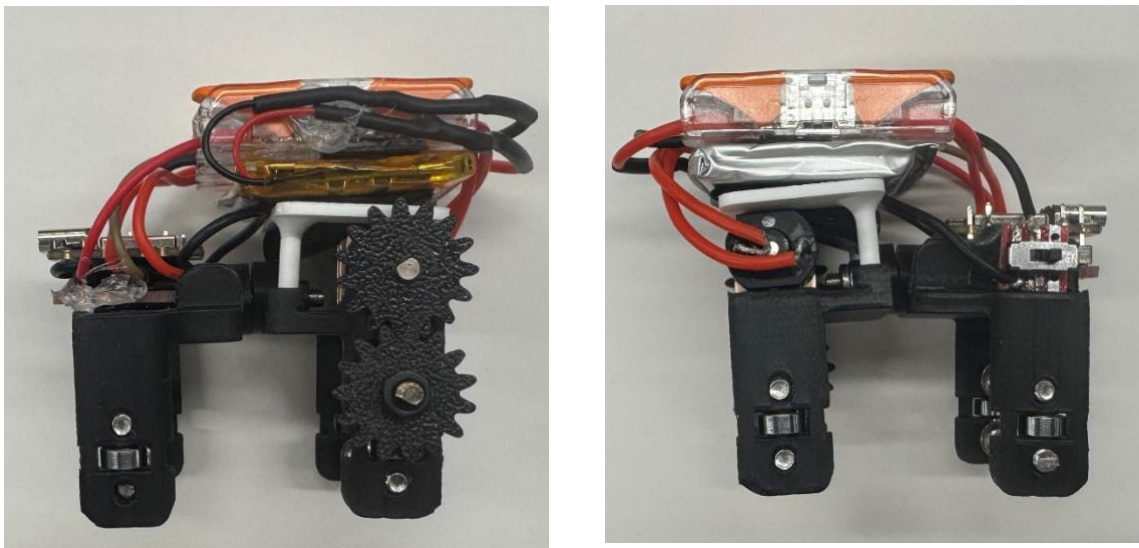


Figure A.2: Left and Right

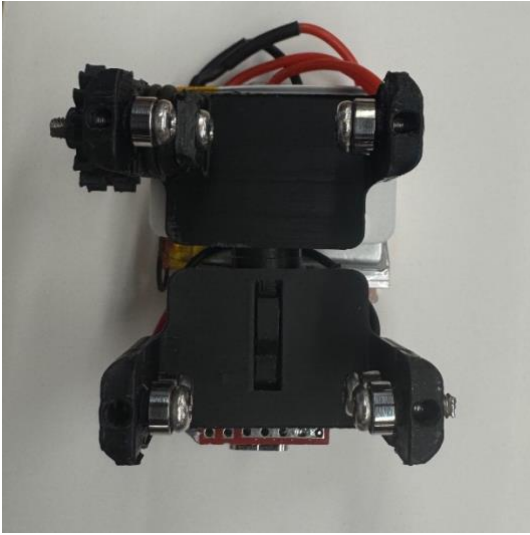


Figure A.3: Top and Bottom

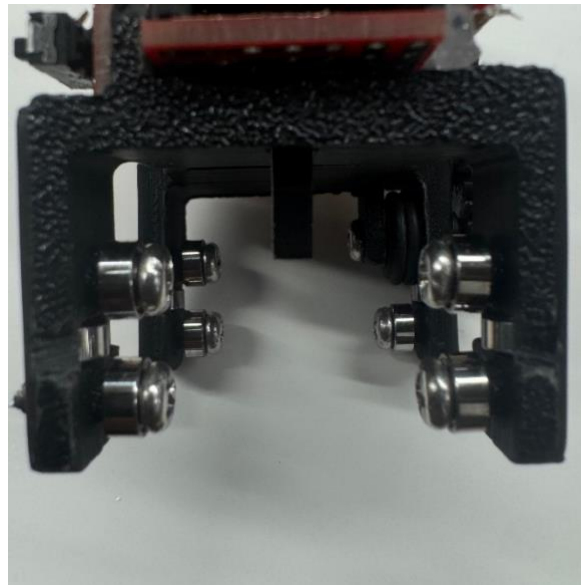


Figure A.4: Wheels Close-Up

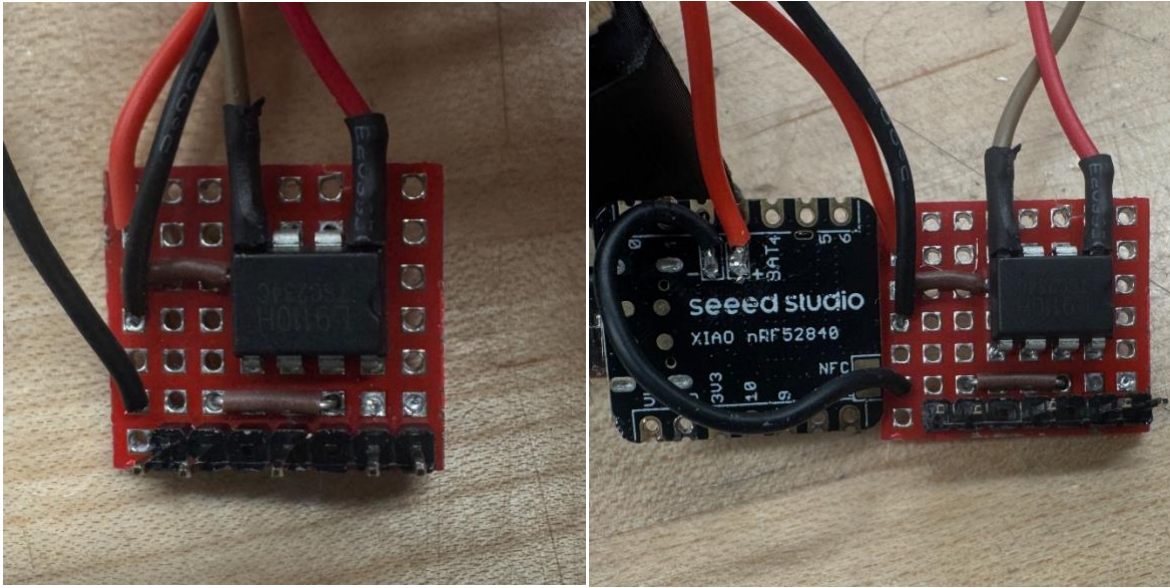


Figure A.5: Wiring Close Up